

## OSSP (Operating Systems And Systems Programming) - 25CS2104E

Roll NO. : 2520030027

Name: Venkata Sai Krishna

SEC-8

### SKILL-1:

**Aim:** To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
venkatsaikrishna@VSK-HP:~$ cd ~
venkatsaikrishna@VSK-HP:~$ mkdir LinuxShellProject
venkatsaikrishna@VSK-HP:~$ cd LinuxShellProject
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ mkdir src include
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ touch Makefile README.md
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ touch src/main.c
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ ls -R
.:
Makefile README.md build docs include src tests
./build:
./docs:
./include:
./src:
main.c
./tests:
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ nano src/main.c
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ nano Makefile
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ make
gcc -Wall src/main.c -o shell
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ ./shell
Linux Shell Project Started!
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ ps
  PID TTY          TIME CMD
   339 pts/0    00:00:00 bash
   633 pts/0    00:00:00 ps
```

```
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ pstree
systemd
├──agetty
├──chronyd-starter─chronyd─chronyd
├──cron
├──dbus-daemon
├──init-systemd(Ub)─SessionLeader─Relay(339)─bash─+
++
├──networkd-dispat
├──rsyslogd─3*[rsyslogd]
├──systemd─(sd-pam)
├──systemd-journal
├──systemd-logind
├──systemd-resolve
├──systemd-udev
├──unattended-upgr─{unattended-upgr}
├──wsl-pro-service─7*[wsl-pro-service]
├──init─{init}
├──login─bash
└──init-systemd(Ub)
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ nano src/fork_demo.c
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ gcc src/fork_demo.c -o fork_demo
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ ./fork_demo
Hello
Hello
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ nano src/exec_demo.c
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ gcc src/exec_demo.c -o exec_demo
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ ./exec_demo
Makefile build exec_demo include src
README.md docs fork_demo shell tests
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ nano src/parent_child.c
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ gcc src/parent_child.c -o parent_child
venkatsaikrishna@VSK-HP:~/LinuxShellProject$ ./parent_child
I am Parent Process
I am Child Process
```

```
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffd95dfef6e0 /* 26 vars */) = 0
brk(NULL) = 0x62d2e39ea000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x779fa2423000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x779fa241e000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779fa23eb000
mmap(0x779fa23f2000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x779fa23f2000
mmap(0x779fa2413000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779fa2413000
mmap(0x779fa241a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x779fa241a000
mmap(0x779fa241c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x779fa241c000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779fa23bd000
mmap(0x779fa23c1000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x779fa23c1000
mmap(0x779fa23e5000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779fa23e5000
mmap(0x779fa23e9000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x779fa23e9000
```

```
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x62d2e3a4d000) = 0x62d2e3a4d000
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_ypi
xel=0}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_ypi
xel=0}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
openat(AT_FDCWD, ".", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY
) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
newfstatat(AT_FDCWD, ".", {st_mode=S_IFDIR|0755, st_size=4096,
..., 0}) = 0
getdents64(3, 0x62d2e3a1e150 /* 13 entries */, 32768) = 384
getdents64(3, 0x62d2e3a1e150 /* 0 entries */, 32768) = 0
close(3) = 0
write(1, "Makefile build exec_demo inc"... , 100Makefile b
uild exec_demo include shell tests
README.md docs fork_demo parent_child src
) = 100
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NU
LL) = 0
munmap(0x779fa2420000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
```

## Code:

```
#include <stdio.h>

int main()
{
    printf("Linux Shell Project Started!\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if(pid == 0)
        printf("I am Child Process\n");
    else
        printf("I am Parent Process\n");

    return 0;
}
```

```
#include <unistd.h>

int main()
{
    execl("/bin/ls", "ls", NULL);
    return 0;
}
```